



Control of actions and tics: A psychophysiological perspective

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Contents

1. Introduction	343
2. Control of actions: Inhibitory control in Tourette syndrome	345
2.1 Psychophysiological correlates of inhibitory control in Tourette syndrome	345
2.2 Inhibitory control in TS: The impact of co-occurring conditions	350
3. Control of tics: Brain correlates of tic suppression	352
3.1 Tic suppression vs resting-state contrasts	352
3.2 Task-based studies of tic suppression	357
3.3 Neurodevelopmental trajectory of tic suppression	357
3.4 Impact of co-occurring conditions on brain correlates of tic suppression	358
4. Conclusions and future directions	358
References	359



1. Introduction

A distinctive feature of Tourette syndrome (TS), compared to other movement disorders, is that in most individuals, tics can be voluntarily suppressed for short periods of time. The fact that one can gain control over their tics is the foundation of an important area of research in TS. Many researchers in the field of TS have pursued efforts to better understand this mechanism, with the ultimate hope that it could lead to improved treatments. Many treatment options for TS, such as exposure and response prevention (Verdellen, Hoogduin, & Keijsers, 2007), as well as Comprehensive Behavioral Intervention for Tics (CBIT; Piacentini et al., 2010), aim to enhance this ability by teaching specific skills. However, it is also possible

that mechanisms involved in improved tic control through behavior therapy differ from those of self-initiated tic suppression. While the brain regions and circuits involved in the generation of tics should be similar across individuals, there might be more than a single way to act on these regions and circuits in order to decrease tic frequency or severity.

In the TS literature, the role played by cognitive control in tic suppression has been the subject of many studies. A recent meta-analysis of 61 cognitive control studies suggested slightly reduced performance in neurocognitive tasks of inhibitory control in individuals with TS (Morand-Beaulieu et al., 2017). As noted in a commentary by Jackson and Jackson (2020), it has been assumed for a long time that the inability to suppress tics stemmed from impaired volitional control or response inhibition mechanisms. However, this remains speculative, and it is not clear that reduced performance during a cognitive control task is associated with the mechanisms involved in tic suppression, or even more so with the mechanisms that give rise to tics in the first place. It is also important to note that ADHD, which co-occurs frequently with TS, contributes significantly to decreased performance in neurocognitive tasks of inhibitory control (Morand-Beaulieu et al., 2017; Sukhodolsky, Landeros-Weisenberger, Scahill, Leckman, & Schultz, 2010).

In parallel, and despite a general tendency for studies to show reduced behavioral performance during inhibitory control tasks, some studies have found enhanced response inhibition capacities among individuals with TS (Jung, Jackson, Parkinson, & Jackson, 2013; Mueller, Jackson, Dhall, Datsopoulos, & Hollis, 2006). It has been suggested that this could represent an adaptive mechanism that emerges in individuals from years of constantly trying to suppress tics. In the same vein, cognitive control has been suggested as one of the putative mechanisms through which behavior therapy may reduce tic severity (Essoe, Ramsey, Singer, Grados, & McGuire, 2021). However, it is still unclear whether cognitive control actually plays a role in CBIT, given that baseline measures of cognitive control failed to predict treatment outcome in large randomized controlled trials of CBIT in children (Chang et al., 2018) and adults (Abramovitch et al., 2017) with TS. Nevertheless, a recent study found a link between a measure of inhibition/switching during the Stroop task and CBIT outcome (McGuire et al., 2021). More work is still required to better understand how cognitive control is related to meaningful tic reduction following behavioral therapy.

While response inhibition/cognitive control tasks offer an interesting insight into control abilities in TS, it is still unclear how they relate to tic

suppression mechanisms. In this chapter, we will review the evidence from psychophysiological studies of response inhibition and tic suppression in order to better understand the similarities and differences in the mechanisms involved in both processes.



2. Control of actions: Inhibitory control in Tourette syndrome

2.1 Psychophysiological correlates of inhibitory control in Tourette syndrome

It is well known that response inhibition involves the prefrontal cortex. Extensive work has shown that the right inferior frontal gyrus is heavily involved in stopping behavior, and is part of a response inhibition network that also includes the pre-supplementary motor area (pre-SMA) and the subthalamic nucleus (Aron, Behrens, Smith, Frank, & Poldrack, 2007; Aron, Durston, et al., 2007; Aron, Robbins, & Poldrack, 2014). In adults with TS, greater involvement of the prefrontal cortex during voluntary inhibition in cognitive control tasks has been reported in many EEG and fMRI studies (Johannes et al., 2001; Morand-Beaulieu, O'Connor, Blanchet, & Lavoie, 2018; Morand-Beaulieu, O'Connor, Sauvé, Blanchet, & Lavoie, 2015; Rae et al., 2020; Serrien, Orth, Evans, Lees, & Brown, 2005; Thibault, O'Connor, Stip, & Lavoie, 2009).

In an early EEG study of response inhibition in TS, Johannes et al. (2001) assessed the ERP correlates of inhibitory control in 10 adults with TS. They assessed an effect known as the “NoGo Anteriorization” (NGA), which reflects the fact that the P3b amplitude has a more frontal distribution during NoGo or Stop trials than during Go trials. They found that the P3b to Stop trials was even more frontally distributed in adults with TS than in controls. A similar effect was observed in three later studies, which all used the same stimulus-response compatibility task that also included a NoGo condition (Morand-Beaulieu et al., 2018, 2015; Thibault et al., 2009). In the first of these three studies, Thibault et al. (2009) observed that the P3b to NoGo stimuli was more frontally shifted in 15 adults with TS compared with 20 controls. However, at this point there were no statistics to support this observation. A few years later, the frontal shift of the NoGo P3b was confirmed and replicated in 20 adults with TS compared with 20 controls (Morand-Beaulieu et al., 2015). A further study added 6 new participants to that TS group and compared it with a different group of

controls and was also able to replicate this effect (Morand-Beaulieu et al., 2018). In the same vein, an early EEG study of TS assessed coherence during a Go/NoGo task in 9 adults with TS and found increased coherence between electrodes spanning the sensorimotor areas and the frontal cortex during the NoGo condition, relative to the Go condition (Serrien et al., 2005). However, this study had a small sample size and assessed sensor-level functional connectivity, increasing the risk of spurious connectivity due to volume conduction. A recent study aimed to circumvent these issues by assessing functional connectivity during a cognitive control task in a large sample of 60 children with TS and by using cortical-source functional connectivity algorithms (Jurgiel et al., 2021). Overall, children with TS exhibited lower spectral power in the ACC, but higher causal information flow between the ACC and other midline central and posterior regions, despite similar levels of task accuracy. Whole brain connectivity analyses highlighted the midcingulate and precuneus as fronto-parietal network hubs whose connections with several motor and sensory areas were atypical in TS. Attenuated ACC activation and greater information flow between and among nodes within the fronto-parietal attention network were significantly associated with tic impairment, faster reaction times, and worse inhibitory performance, suggesting it may be a putative neural adaption among individuals with TS.

Studies using fMRI to assess the brain correlates of cognitive control have suggested similar findings. For instance, Marsh, Zhu, Wang, Skudlarski, and Peterson (2007) examined neural activation in regulatory regions during the Stroop test in children and adults with and without TS and looked at effects of age and Stroop test performance on brain activation. Stroop test performance was associated with different areas of the brain in TS and controls. In TS, poorer performance was associated with greater activation of frontostriatal regions, whereas in controls, more activation of the right inferolateral prefrontal cortex accompanied better performance. The two groups also differed in their correlation of age with magnitude of brain activation during the task, specifically in regions associated with the default mode network: TS showed decreased activation in the ventral prefrontal and posterior cingulate cortices with increased age. These results suggest that individuals with TS show greater difficulty in actively suppressing neural activity/regulating during the more active task. Additionally, they struggle to engage mental processes that become active when freed from the demands of more challenging cognitive tasks. Furthermore, since Stroop test performance was unimpacted in TS, there may be a compensatory

process occurring where greater activation of the frontostriatal systems helps individuals with TS maintain task performance.

Two further studies showed increased activation of the right prefrontal cortex in children ([Jackson et al., 2011](#)) and adults ([Rae et al., 2020](#)) with TS during tasks involving cognitive control. [Jackson et al. \(2011\)](#) assessed the functional and structural correlates of cognitive control during a manual task-switching paradigm in children with TS and typically developing controls. Enhanced cognitive control was found in children with TS during the task. The authors then used diffusion-weighted imaging (DWI) to see whether this enhanced cognitive control was accompanied by white matter changes in TS. Compared with typically developing controls, children with TS had reduced fractional anisotropy and increased mean diffusivity in the corpus callosum and forceps minor, which connects the lateral and medial areas of the prefrontal cortex. Further, this white matter microstructure was associated with tic severity. The authors believe changes in cortical white matter most likely reflect a functional adaptation to TS rather than a core symptom of the disorder. But it is possible that white matter differences could alternatively reflect individual variation in connectivity that might confer on some individuals with TS an increased capacity to suppress tics. Performance on the behavioral task (reaction time) was strongly associated with the white matter of corpus callosum in typically developing controls but not in children with TS. In the TS group, reaction times were associated with the white matter microstructure of the forceps minor instead. This suggests that for the TS group, the normal role performed by the corpus callosum in resolving the selection of motor outputs may be reduced in favor of an increased role for the prefrontal cortex. The experimenters followed up DWI analyses with fMRI to see if there were significant differences in brain activation during the manual task-switching paradigm. Compared with typically developing controls, children with TS had greater brain activation in the right prefrontal cortex during high conflict incongruent-switch trials relative to incongruent-repeat trials. In the TS group only, the fMRI BOLD response was very strongly and positively linearly associated with RT. Authors concluded that for the TS group only, enhanced cognitive control of motor output is accompanied by adaptive functional and structural changes in the prefrontal cortex. [Rae et al. \(2020\)](#) used a modified Go/NoGo task to uncover neural mechanisms by which adults with TS control actions. The researchers reported no group differences in task accuracy or reaction times between adults with TS and age-matched controls. They also reported that the neural processes by which

adults with TS choose to act upon or withhold movements was similar to that of controls, as activity in the cardinal prefrontal and motor regions (including preSMA area) was anatomically similar in both groups. However, adults with TS demonstrated greater activity in the right inferior frontal gyrus across all task conditions.

Another inhibitory control study also highlighted the role played by the right prefrontal cortex in TS. It was found that in medicated adults with TS, gray matter signal in the right inferior frontal gyrus was correlated with stop-signal reaction times in an emotional stop-signal task (Atkinson-Clement et al., 2020). Stop signal reaction times were also associated with higher connectivity between the inferior frontal gyrus and the superior temporal gyrus, and with lower connectivity between the inferior frontal gyrus and the cerebellum. While these results were restricted to those on medication, it is possible that these individuals had increased tic severity (before medication), such that they would represent more severe cases of TS than unmedicated individuals.

Whereas this pattern of frontal overactivation has been reported across many studies using a wide range of experimental paradigms, it is important to note that some studies did not find such frontal overactivation associated with inhibitory control in TS. Notably, Debes, Hansen, Skov, and Larsson (2011) found no difference in brain activation between children with TS and typically developing controls during a Stroop or a Go/NoGo task. Also, Thomalla et al. (2014) only reported decreased brain activation in the left primary motor cortex and secondary motor areas during movement execution (Go trials) but not during response inhibition (NoGo trials) in adults with TS. Another study found increased right inferior parietal cortex activation in adults with TS relative to controls, but no difference over prefrontal regions (Fan et al., 2018). While many EEG studies reported frontal overactivation in the P3b time window for NoGo stimuli (Johannes et al., 2001; Morand-Beaulieu et al., 2018, 2015; Thibault et al., 2009), two recent studies using the stop-signal (Schüller et al., 2018) or the stop-change task (Brandt, Stock, Munchau, & Beste, 2017) reported decreased frontal P3b in adults with TS. Yet, in the study by Schüller et al. (2018), a decreased frontal P3b was observed across condition and was not specific to the stop condition.

In TS studies that reported a frontal overactivation associated with inhibitory control, it has often been interpreted as the source of inhibitory control of motor outputs, suggesting that it may also play a role in the control of tics. However, it is not clear whether this brain pattern is indeed adaptive.

For instance, it was found that the NoGo P300 over frontal electrodes was negatively correlated with reductions in tic severity after cognitive-physiological (CoPs) therapy, and that it was also positively correlated with endpoint tic severity (Morand-Beaulieu, 2019, 2018). This suggests that adults with TS presenting with increased frontal activation during response inhibition respond less to this type of treatment and continue to show elevated tic severity after therapy. A study from Jung et al. (2013) argues that the role played by the prefrontal cortex in TS is not one of inhibitory control of motor outputs, but instead, that it is the source of facilitatory signals that may operate to bias response competition downstream within brain areas more directly linked to motor planning or motor execution. In support of this theory, authors showed that for children with TS, increased task difficulty was associated with increased activation of the inferior frontal cortex and the supplementary motor area. Tic severity was also positively associated with brain activation in these two regions, as well as with activation in the primary motor cortex and the anterior cingulate cortex. Thus, results run counter to the general idea that increased prefrontal cortex activation leads to increased inhibition of motor responses but is more consistent with recent reports that regions such as supplementary motor area may bias suppression mechanisms elsewhere instead of being the direct source of suppression.

In addition to the study of Jung et al. (2013), other studies have highlighted the important role of the supplementary motor area in inhibitory processes in TS. Ganos et al. (2014) conducted an fMRI study of inhibitory control in adults with TS using the stop-signal task. They found that tic frequency was positively correlated with activations in supplementary motor area during the Stop vs Go contrasts. In agreement with Jung et al. (2013), they concluded that the supplementary motor area seems to act as a global inhibition mechanism in TS and might be used to simultaneously stop tics and voluntary actions. Another study that also used a stop-signal task similarly reported worse cognitive control performance in medicated adults with TS to be associated with increased gray matter signal in the supplementary motor area (Atkinson-Clement et al., 2020). An EEG study also showed that lateralized readiness potentials (LRP), which are partly generated by the supplementary motor area, had an increased amplitude in adults with TS relative to controls during a stimulus-response compatibility task (Morand-Beaulieu et al., 2018). This increased LRP amplitude was also correlated with the severity of phonic tics. Finally, it was found that error-related brain activity in the supplementary motor area was increased during a cognitive control task in adults with TS (Fan et al., 2018).

2.2 Inhibitory control in TS: The impact of co-occurring conditions

Co-occurring conditions are common in TS. It is said that up to 90% of individuals with TS also present with a co-occurring condition, the most frequent being ADHD and OCD (Bitsko et al., 2014; Freeman et al., 2000; Hirschtritt et al., 2015). Co-occurring ADHD is known to have an impact on inhibitory control in TS (Morand-Beaulieu et al., 2017) and could, thus, explain some inconsistencies between studies. The best way to disentangle the effects of ADHD (or any other co-occurring condition) is by using a 2×2 factorial design, where we have a group with TS-only, a group with ADHD-only, a group with both conditions (TS + ADHD) and a group with neither (controls) (Rothenberger & Heinrich, 2021). This design was used in an ERP study that assessed how TS and ADHD may act separately or additively on electrophysiological markers of inhibitory controls (Shephard, Jackson, & Groom, 2016). In this study, ERPs were recorded while children with TS-only, ADHD-only, TS + ADHD, and typically developing controls performed a Go/NoGo task. Analyses revealed that reduced P3b amplitude was associated with the ADHD factor, rather than the TS factor, suggesting that children with ADHD-only and TS + ADHD had a smaller P3b amplitude than those with TS-only and typically developing controls. Recently, our research team aimed to expand this previous work by including a larger sample of children with TS and/or ADHD, and by adding time-frequency analyses on top of ERPs (Morand-Beaulieu et al., 2022). Similar to the findings of Shephard et al. (2016), we found that children with ADHD-only had reduced task performance and frontal mid-line theta, as well as delayed latency of N2 and P3 ERPs, while those with TS-only did not differ from typically developing controls on any measures. However, we found unexpected results regarding children with TS + ADHD. Contrary to our expectations and to the findings of Shephard et al. (2016), children in this group showed similar EEG markers of inhibitory control to that of typically developing controls and did not significantly differ from them in terms of task performance either. It is possible that this difference could be explained by the severity of tic symptoms across studies. In our study, the group of children with TS + ADHD did not differ from the group of children with TS-only in terms of tic severity as assessed with the YGTSS. However, in the study of Shephard et al. (2016), children with TS + ADHD not only had more ADHD symptoms than children with TS-only, but they also had increased TS symptoms. This difference in symptoms between studies may in part explain the observed difference in EEG markers

between them. It is also possible that some ADHD symptoms in children with TS could be related to TS-associated factors, such as reduced attention due to attempts at suppressing tics or by being distracted by one's own tics (Erenberg, 2005).

Eichele et al. (2016, 2017) also conducted an ERP study comparing children with TS, ADHD, and typically developing controls. In this study, they assessed the development of cognitive control using a modified Eriksen flanker task at two assessments separated by 5 years. At the first assessment, TS showed higher amplitudes of an early P3 component across experimental conditions and in separate trial outcomes. However, the early P3 showed a significant increase for all groups, suggesting that it develops earlier in children with TS compared with controls, but later converges over development. While the TS group was small ($n = 17$), the authors still performed subgroup analyses to compare children with TS-only ($n = 7$) to those with TS + ADHD ($n = 10$). Similar to our findings, the authors did not find major differences between these subgroups on behavioral or electrophysiological measures of cognitive control.

However, an fMRI study has reported differences between children with TS without ADHD (TS-ADHD) and those with TS + ADHD during a stop-signal task (Openneer et al., 2021). This study reported increased stop-signal reaction times during the task, as well as atypical neural patterns during failed inhibition in the TS + ADHD group but not in the TS-ADHD group, indicating that co-occurring ADHD was the driving force behind these differences. For all children, neural activation was found in areas associated with response inhibition, including the inferior and prefrontal gyrus, insula, and temporal gyrus. However, only in children with TS + ADHD was increased activation found in the right inferior frontal gyrus and the left insula during failed inhibition. This overactivation could represent either the effects of ADHD symptoms in TS during difficult conditions or the combined effects of controlling tics and associated premonitory urges. Increased activity of the middle & superior temporal gyrus was also found for children with TS + ADHD during failed inhibition, which could underscore the possibility of an immature inhibitory network in children with TS + ADHD, leading to cognitive difficulties.

Many factors could explain these discrepancies between studies, such as the level of cognitive demand implied by different tasks, the age of participants, or the severity of TS and/or ADHD symptoms. In this vein, the study of Rae et al. (2020) revealed that greater ADHD severity was associated with greater connectivity between the inferior frontal gyrus & premotor cortex

during Choose-NoGo trials. This suggests that adults with TS with worse ADHD symptoms require stronger leverage of prefrontal resources to motor cortices in order to withhold actions.



3. Control of tics: Brain correlates of tic suppression

3.1 Tic suppression vs resting-state contrasts

Most individuals with TS are able to suppress their tics for short periods of time. However, to the best of our knowledge, there have only been eight studies to date that tried to understand the physiological processes involved in tic suppression. The first such study was conducted by [Peterson, Skudlarski, Anderson, et al. \(1998\)](#), who assessed the brain correlates of tic suppression in 22 adults with TS using fMRI. In contrast with periods where they were allowed to tic freely, adults with TS showed increased and decreased activations in various areas of the brain during tic suppression. In most subcortical areas (bilateral putamen, thalamus, and globus pallidus), there were reduced activations during tic suppression. The right caudate nucleus was the only subcortical structure showing increased activation during tic suppression. At the cortical level, there were increased brain activations in the right frontal cortex, bilateral temporal gyri, right anterior cingulate cortex, and bilateral occipital cortices. There were also decreased activations in the right posterior cingulate cortex, the bilateral hippocampus and cuneus, as well as the left sensorimotor and inferior parietal cortices. Correlational analyses also showed that baseline tic severity (outside of the scanner), was negatively associated with increased/decreased activations of subcortical structures (as well as sensorimotor and inferior parietal cortices), such that participants showing the most important changes in activation during the tic suppression task were those with the least severe tics. While this study showed that many brain structures were activated/deactivated during tic suppression, the increased activation of the right frontal cortex has been used for a long time to support the relationship between tic suppression and inhibitory control mechanisms. Frontal cortex involvement in tic suppression is also supported by findings of increased regional homogeneity in the left inferior frontal gyrus during tic suppression ([Ganos et al., 2014](#)). While the right inferior frontal gyrus is known to play a major role in inhibitory control, as one of the key nodes of the stop network ([Aron et al., 2014](#)), the role played by the left inferior frontal gyrus in inhibition is less clear.

Another study, which was conducted by [Serrien et al. \(2005\)](#), is also often referenced to support the parallel between inhibitory control and tic suppression. In this study, coherence analyses were conducted to assess the functional connectivity between pairs of EEG electrodes while 9 adults suppressed their tics. The authors found that EEG alpha coherence between all electrode pairs was increased during tic suppression. Notably, there were four connections (F3-C3, FCz-C3, FCz-C4, F4-C4) where coherence was more significantly increased during tic suppression relative to rest. According to the authors, these connections represented increased functional connectivity between the sensorimotor areas and the prefrontal and mesial frontal cortex. As mentioned earlier in this chapter, the authors also assessed the same connections during a Go/NoGo task and found that coherence was increased during the NoGo in adults with TS. This led to the conclusion that response inhibition and tic suppression may occur through the same fronto-mesial network. Another study also performed similar analyses, but in 9 children with Tourette syndrome ([Hong et al., 2013](#)). Here, the authors used partial directed coherence to analyze effective connectivity between 18 scalp electrodes. Authors compared tic suppression and rest for each possible connection (306 connections) and five frequency bands (delta, theta, alpha, beta, gamma) using paired-sample *t*-tests and found increased coherence in the theta band from electrodes F3 and C3 to T4. In the beta band, they reported increased coherence from F3 and F8 to T3. However, these results did not survive correction for multiple comparisons and thus their meaningfulness may be limited. Additionally, they found increased delta power at F7 and increased theta power at Fp1 during tic suppression, but these results did not survive correction for multiple comparisons either. While these two latter studies reported increased connectivity during tic suppression compared to rest, they both share methodological shortcomings including small samples and limited spatial resolution. The latter issue is mostly attributable to the small number of electrodes used in these studies and to problems associated with volume conduction when assessing connectivity at the sensor level. It is, therefore, complicated to interpret the brain structures involved in the connection highlighted in both studies.

We tackled these issues in one of our recent studies ([Morand-Beaulieu et al., 2021](#)), where we aimed to replicate and extend the earlier findings of [Serrien et al. \(2005\)](#) of increased alpha-band connectivity during tic suppression. While their study was conducted in 9 adults with TS, we recruited a large sample of 72 children with TS. Additionally, we assessed functional

connectivity during tic suppression on source-reconstructed EEG, a technique that overcomes some limitations of sensor-level functional connectivity, such as the risk of spurious connectivity due to volume conduction and the limited capacity for consideration of the brain regions involved. Our study revealed increased alpha-band functional connectivity during tic suppression compared with a resting-state when tics were not suppressed. Our finding of increased connectivity in the alpha band is consistent to the seminal findings of [Serrien et al. \(2005\)](#). However, our approach allowed for a better refinement of the brain regions involved in these connections and consequently suggested potentially different conclusions. Our analyses revealed a subnetwork of 28 nodes that were connected through 29 edges. Within that subnetwork, there were three hubs, i.e., brain regions that were connected to many others in the same subnetwork. These hubs were the right superior frontal gyrus, the left precuneus, and the left banks of the superior temporal sulcus ([Fig. 1](#)).

In the Desikan–Killiany atlas that we used for brain parcellations, the anatomically-defined superior frontal gyrus encompasses the functionally-defined dorsomedial prefrontal cortex ([Desikan et al., 2006](#)). Both the SFG and the precuneus have been identified by previous fMRI investigations as part of the default mode network ([Buckner, Andrews-Hanna, & Schacter, 2008](#); [Utevsky, Smith, & Huettel, 2014](#)), suggesting that this large-scale brain network may be involved in voluntary tic suppression. It may also suggest that brain mechanisms involved in tic suppression differ from those of response inhibition. This hypothesis is supported by a recent fMRI study of tic suppression, which contrasted tic suppression in adults with TS to eyeblink inhibition in controls ([van der Salm et al., 2018](#)). In controls, eyeblink inhibition mainly activated the sensorimotor circuit, including notably the right ventrolateral prefrontal cortex, the premotor cortex, and the supplementary motor area. This circuit is typically involved in response inhibition. In adults with TS, tic suppression mainly activated the superior prefrontal gyrus, the anterior and dorsolateral prefrontal cortex, and the dorsal anterior cingulate cortex. Based on these findings, the authors' interpretation was that in controls, the "classical stop network" is recruited to inhibit eyeblinks, while individuals with TS recruits the "limbic network" to suppress tics. This latter study beautifully demonstrated that voluntary tic suppression could involve a mechanism that is distinct from that of response inhibition.

Despite different approach and interpretations, these results may be consistent with those of our study of source-EEG connectivity during tic

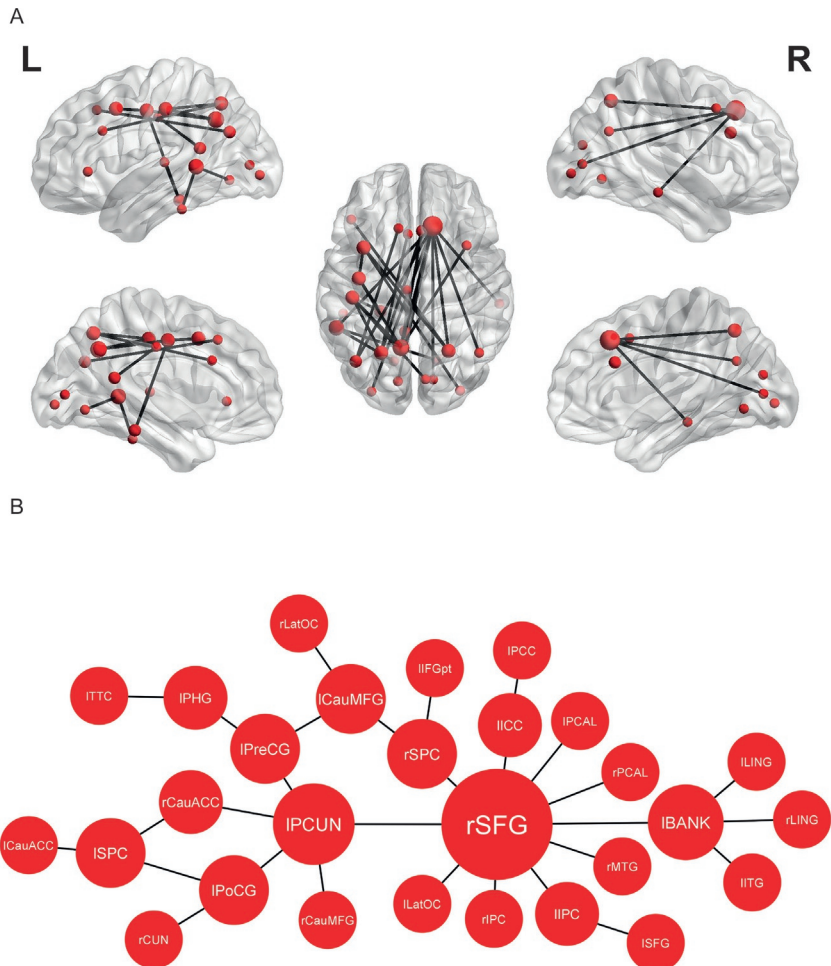


Fig. 1 Subnetwork of functional connectivity in the alpha band during tic suppression. Nodes depicted in this figure showed increased connectivity during tic suppression in comparison to rest. Node size corresponds to connectivity degree (number of connections per node). (A) Hubs in this network were the right superior frontal gyrus (10 connections, degree z-score: 4.2), the left precuneus (5 connections, degree z-score: 1.5), and the left banks of superior temporal sulcus (4 connections, degree z-score: 1.01). (B) Graphical representation of the alpha-band subnetwork. Note: “l” & “r” denotes the lateralization of the nodes; CauACC: caudal anterior cingulate cortex, CauMFG: caudal middle frontal gyrus, CUN: cuneus, ICC: isthmus of the cingulate cortex, IFGpt: pars triangularis of the inferior frontal gyrus, IPC: inferior parietal cortex, ITG: inferior temporal gyrus, LatOC: lateral occipital cortex, LING: lingual gyrus, MTG: middle temporal gyrus, PCAL: pericalcarine cortex, PCC: posterior cingulate cortex, PCUN: precuneus, PHG: parahippocampal gyrus, PoCG: postcentral gyrus, PreCG: precentral gyrus, SFG: superior frontal gyrus, SPC: superior parietal cortex, TTC: transverse temporal cortex. *Reproduced from Morand-Beaulieu, S., Wu, J., Mayes, L.C., Grantz, H., Leckman, J.F., Crowley, M.J., Sukhodolsky, D.G., 2021. Increased alpha-band connectivity during tic suppression in children with Tourette syndrome revealed by source EEG analyses. Biological Psychiatry: Cognitive Neuroscience and Neuroimaging.*

suppression in children. [van der Salm et al. \(2018\)](#) argued that the pattern of activations during tic suppression suggested that it relied on the limbic network. However, the existence of the limbic network has been a subject of debate in the neuroscience literature in the recent years ([LeDoux, 2012](#); [Pessoa & Hof, 2015](#)). In a recent review paper, [Uddin, Yeo, and Spreng \(2019\)](#) proposed a taxonomy of large-scale brain networks that is based mostly on neuroanatomical rather than functional labels. For instance, networks known as the DMN or the fronto-parietal control network in the cognitive domain are, respectively, named the medial and lateral frontoparietal networks in this taxonomy. Furthermore, in the Uddin classification, this limbic network is subsumed by the medial frontoparietal network (or DMN). Therefore, this classification appears to be the key to connect our results with those of [van der Salm et al. \(2018\)](#). In their study, the brain regions activated during tic suppression were mostly midline frontal areas. These regions also somewhat overlapped with one of the core hubs of increased connectivity during tic suppression in our study in children with TS. Additionally, in the study from [van der Salm et al. \(2018\)](#), brain regions activated during eyeblink inhibition in healthy controls are consistent with those that are included in the lateral frontoparietal network. Indeed, the inferior frontal gyrus/ventrolateral prefrontal cortex, which was one of the largest clusters activated during eyeblink inhibition, is a core region of the lateral frontoparietal network.

Why increased DMN connectivity would occur during tic suppression remains to be elucidated. It is possible that individuals with TS engage in self-referential thinking while they are instructed to suppress their tics. Self-referential thinking has been shown to involve the DMN ([Cavanna & Trimble, 2006](#); [Gusnard, Akbudak, Shulman, & Raichle, 2001](#); [Sheline et al., 2009](#)). In this vein, a recent study reported increased connectivity between the DMN and basal ganglia and hypothesized that it could represent a brain correlate of self-referential thinking related to tics ([Ramkiran, Heidemeyer, Gaebler, Shah, & Neuner, 2019](#)). Thus, children in our study might have been more focused on premonitory urges and incoming tics, relying on self-referential processing to suppress tics, which would result in increased DMN connectivity.

Additionally, the sensorimotor network (SMN), which stands in a reciprocal relationship with the DMN ([Martino et al., 2016](#); [Northoff, Hirjak, Wolf, Magioncalda, & Martino, 2021](#)) is known to be involved in tic generation ([Bohlhalter et al., 2006](#); [Neuner et al., 2014](#)). This reciprocal relationship means that when activity in the DMN goes up, activity in

the SMN goes down, and vice versa. Thus, increased DMN connectivity during tic suppression could balance and reduced hyperactive sensorimotor network activity, resulting in reduced tic frequency.

3.2 Task-based studies of tic suppression

While all of these previous studies contrasted two “resting state” conditions—one where tics were suppressed and one where participants were allowed to tic freely—a recent EEG study assessed tic suppression during a motor task (Zapparoli, Macerollo, Joyce, Martino, & Kilner, 2019). This study showed that while adults with TS were ticcing freely, they did not show the same modulation in EEG beta power that was observed in controls during finger movement (decrease in beta power). However, when they were asked to suppress their tics, their decrease in beta power during finger movement was larger than during the tic freely condition. In other words, the modulation of beta oscillations during finger movement during tic suppression was similar to that of controls. According to the authors, this suggests that tic suppression operates by normalizing abnormal beta oscillations over the sensorimotor cortex, at least during the context of a finger movement task.

However, it is possible that individuals with TS use different strategies to suppress tics in different contexts, and thus different mechanisms may be involved in tic suppression depending on context. In studies where individuals with TS are asked to suppress tics during a resting-state session, there are no other instructions than to suppress tics. Therefore, participants may either be intensively focused on their tics and premonitory urges or may let their mind wander. These experimentations may reflect some real-life situations such as sitting at home or in a waiting room, where no other actions are expected or required. Conversely, the experimental manipulation in the study of Zapparoli et al. (2019) may reflect what happens in a classroom or at work, when tics may be suppressed while other tasks are performed. Therefore, future studies aiming to understand the neural correlates of tics suppression should assess these in various contexts and tasks that are close proxies to the everyday life of individuals with TS.

3.3 Neurodevelopmental trajectory of tic suppression

Previous studies have shown that tic suppression follows a developmental trajectory. In a pooled analysis of 9 behavioral studies of tic suppression, Conelea et al. (2018) found that tic suppressibility was positively correlated

with age. Only two studies have assessed the brain correlates of tic suppression in children (Hong et al., 2013; Morand-Beaulieu et al., 2021), thus making it hard to understand how brain development may impact tic suppression. In our recent study, we found that connectivity within a specific subnetwork, in which the right superior frontal gyrus was the main hub, was positively correlated with age during tic suppression but not during rest (Morand-Beaulieu et al., 2021). Therefore, in that subnetwork, connectivity did not differ between rest and tic suppression in younger children, but older children and adolescents showed increased connectivity relative to rest. However, it remains unclear whether this increased connectivity in older children and adolescents plays a role in tic suppression capacities.

3.4 Impact of co-occurring conditions on brain correlates of tic suppression

Behavioral studies of tic suppression suggest that co-occurring conditions do not impact tic suppression. Despite two small studies suggesting a possible negative impact of attention problems on tic suppression (Himle & Woods, 2005; Müller-Vahl, Riemann, & Bokemeyer, 2014), a recent large study reported no impact of ADHD on tic suppressibility (Conelea et al., 2018). Additionally, the latter study reported no impact of OCD or anxiety disorders as well. Among the few psychophysiological studies that have assessed the impact of co-occurring conditions on brain correlates of tic suppression, none have reported a significant effect of either ADHD or OCD (Morand-Beaulieu et al., 2021; Peterson et al., 1998).



4. Conclusions and future directions

In this chapter, we reviewed the psychophysiological literature on response inhibition and tic suppression in children and adults with TS. While seminal psychophysiological studies of tic suppression suggested similar mechanisms for response inhibition and tic suppression, more recent studies revealed that these processes are probably more different than previously hypothesized. However, tic suppression remains understudied and poorly understood. It is something that most individuals with TS do on an almost-daily basis but how it occurs remains somewhat unclear. It is an interesting and intriguing feature of TS and something that distinguishes it from most other movement disorders. A better understanding of this phenomenon may potentially allow us to enhance tic suppression capacity, such as through behavioral treatments.

One way to further our understanding of tic suppression would be to develop new paradigms to study it. Current paradigms that have been used in many psychophysiological studies, such as a contrast between tic suppression and a “resting-state,” may lack ecological validity. For instance, it would be interesting to contrast brain activity in settings where tics are more likely to be suppressed, such as the classroom or the workplace, and settings where active tic suppression is less likely, such as being home alone.

Furthermore, future psychophysiological studies of tic suppression should recruit larger and more diverse samples. Current evidence does not show a gender effect (Conelea et al., 2018), but the number of girls and women recruited in tic suppression studies is likely too small to identify any potential gender and/or sex effects. Understanding how neuro-development affects tic suppression is also very important and future studies should additionally aim to assess tic suppression at different developmental stages.

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